

# ROBUST VARIATIONAL INFERENCE VIA IMPRECISE PROBABILITIES

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# Contents

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## **List of Tables**

## List of Figures

# Abstract

## ROBUST VARIATIONAL INFERENCE VIA IMPRECISE PROBABILITIES

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The aim of the thesis is to investigate the performance of Gismos and to design and construct a super multi-functional Gismo.

The novel aspects of the new Gismo are described. The abstract should perhaps be about half a page long.

The results of testing, which show the abject failure of the Gismo, are presented.

In the conclusions proposals for rectifying the deficiencies are outlined.

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I would like to thank...



# Chapter 1

## Introduction

### 1.1 Background and motivation

Bayesian inference provides a rigorous framework for updating beliefs about model parameters in the light of observed data, yet the uncertainty it quantifies is not immune to uncertainty itself. This is the problem of second-order uncertainty, which arises because Bayesian inference carries a foundational assumption that the prior distribution is known and correctly specified. When multiple priors are equally plausible, committing to a single one is epistemically insufficient and can lead to overconfident inferences. The framework is precise about uncertainty over parameters but it fails to capture the uncertainty about the choice of the prior which precedes inference at the level of modelling itself. This problem has been recognised in the philosophy of probability since at least the 1960s, formalised under the study of imprecise probabilities and credal sets, yet it has remained largely intractable to address in modern probabilistic machine learning. The availability of large datasets and powerful computational resources has led to a focus on the scalability of inference algorithms, but the problem of second-order uncertainty has been largely neglected.

Prior sensitivity is a routine challenge in Bayesian inference, and when the data is scarce, the likelihood is not sufficiently informative to overwhelm the influence of the prior and posterior remains sensitive to the initial modelling choices. In settings such as healthcare, ecological modelling or scientific parameter estimation, data is expensive to obtain or just inherently limited; two practitioners using equally defensible priors can arrive at substantially different posteriors and therefore different conclusions.

Variational Inference (VI) has emerged as a powerful tool for Bayesian computation, offering a scalable alternative to Markov Chain Monte Carlo (MCMC) methods. However, standard VI inference inherits the same foundational limitations of Bayesian inference, and it is not designed to account for the uncertainty about the choice of the prior. Credal sets offer a natural remedy by allowing us to consider sets of priors instead of a single one, we obtain a corresponding set of posteriors, and the inferential question shifts from finding the best approximation to a single target to asking what can be concluded robustly across the entire set. But the computational challenges of working with sets of distributions are formidable.

Recent advances in normalising flows and their continuous-time generalisation through neural ordinary differential equations (Chen et al., 2018) have made expressive posterior approximation tractable. Flow matching (Lipman et al., 2022) represents a further step, replacing the computationally expensive log-determinant evaluations required by classical normalising flows with a direct regression objective over a learned velocity field, making training a shared posterior approximation across multiple priors substantially more scalable.

This thesis investigates whether credal mixtures of flow-based posteriors can provide robust inference under prior uncertainty. The approach is grounded in a classical result from information theory: the KL divergence is convex in its second argument, which guarantees in principle that a mixture over a set of posteriors performs at least as well as the best individual component (Cover and Thomas, 2006). Whether this guarantee remains empirically visible under deep variational approximation, and what architectural and methodological conditions are required for it to hold, is the central question this work addresses.

## 1.2 Problem statement and objectives

The central problem this thesis addresses is the gap between the theoretical promise of credal mixture inference and their empirical behaviour under deep variational approximation. While the convexity of the KL divergence guarantees that a mixture over a set of posteriors should perform at least as well as the best individual component, it is not clear whether this guarantee remains empirically visible when using deep neural networks to approximate the posteriors. These approximations are learnt under finite

data regime and the mixture weights must be estimated from the held-out data. These sources of approximation and estimation error may undermine the theoretical guarantee, and it is an open question under what conditions the robust performance of credal mixtures can be realised in practice.

More concretely, the problem has two interacting components. The first is architectural: if each posterior approximation is trained independently under a different prior, there is no guarantee that the resulting approximations are comparable in quality or that their mixture is well-defined as a density. The second is statistical: even with a coherent family of approximations, estimating mixture weights using proxy objectives such as maximum mean discrepancy or adversarial divergence bounds introduces estimation noise that can cause the mixture to underperform its best component. Both failure modes were observed in preliminary experiments and motivate the specific design choices investigated in this thesis.

The objectives of this work are fourfold. First, to design a shared-backbone flow matching architecture conditioned on a discrete prior index, ensuring that all posterior approximations are trained within a common optimisation framework with comparable capacity. Second, to investigate likelihood-consistent mixture weight selection via held-out negative log likelihood as a principled alternative to proxy divergence objectives such as maximum mean discrepancy and adversarial KL bounds, which were found in preliminary experiments to introduce estimation noise sufficient to mask the theoretical guarantee. Third, to evaluate whether the resulting credal mixtures recover the convexity guarantee empirically, using paired per-seed comparisons to control for estimation variance. Fourth, to characterise the conditions under which the theoretical guarantee is and is not empirically visible, contributing to a broader understanding of when classical information-theoretic arguments remain reliable under deep variational approximation.